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A Topological Tidal-Curvature Imager: Calorimetric Readout of Edge-State Response in Curved-Spacetime SSH Arrays

Draft for comment — not peer-reviewed. Companion: *A Placement Law for Shift Sensing in Curved-Spacetime SSH Chains* (shift-law-draft).

Status note (17 Oct 2026). A cold review (repository `review/`) established that the far-end lead of section 3 does not set the probed mode's linewidth — the mode's weight at the lead end is 10^{-151} — so every absolute-power figure in section 5 rests on a linewidth this draft never specifies, and the dissipative channel must be redesigned at the probed end before those figures mean anything. The placement law and all responsivity results of section 4 are unaffected (they are linewidth-independent) and have been independently reproduced. Section 5 is retained as written, under this caveat, pending that redesign; an on-site-disorder budget for the device is also owed.

Authorship note. This work was developed in extended collaboration between B. Watson (direction, framing, verification standard) and an AI system (Claude, Anthropic), which performed the bulk of the formal derivation, numerics, and drafting under the author's direction. The companion paper's two-session cross-verification protocol was applied retroactively to this paper's central law in August 2026.

Abstract

We propose a curvature imager built from the curved-spacetime Su-Schrieffer-Heeger model: a dimerized chain whose position-dependent hopping encodes an engineered metric, read by a weakly coupled qubit whose decoherence rate is measured calorimetrically through the power dissipated in maintaining the chain's protected edge mode. Our central result is a placement law governing the readout's sensitivity: the logarithmic response of the edge density of states to curvature equals the differential metric tilt integrated over the mode's own probability footprint. Two exact corollaries follow—uniform metric changes cancel identically for the protected mode, and probe placement becomes a one-line calculation from the bond profile. The sensitive end of a warped chain is the horizon end, where the local velocity vanishes; the far end is quiet by construction, its responsivity falling as $1/(3N)$. The law carries a falsifiable signature: a responsivity asymmetry between the two ends of one warped chain — a factor of 151 at the 250-cell reference chain, growing linearly with chain length — testable in principle in room-temperature photonic platforms. A worked superconducting implementation, with every parameter derived here, cited to demonstrated hardware, or stated explicitly as a design choice within demonstrated ranges, yields signal-to-noise of 2 at one hundred seconds and 22 at three hours for a change $\delta\sigma = 0.01$ in the warping exponent, detector-limited by demonstrated zeptowatt-scale bolometry (figures under revision — see status note). The instrument does not compete with optical clocks on absolute dilation and is confined by its physics to engineered and analog metrics, where it targets what point measurements cannot provide: a spatially resolved image of how a laboratory metric varies.

1. Introduction

This work began as a search for novel sensing architectures for spacetime curvature—instruments that could complement, rather than compete with, the point-precision of optical atomic clocks. Clocks now resolve gravitational redshift across millimeters, but they measure the metric at a point. What they do not provide is a dense, spatially resolved image of how curvature varies across a region. That imaging gap is the niche this instrument targets, and we state its domain honestly at the outset: the signal magnitudes accessible to this architecture place it in the regime of engineered and analog metrics—strained lattices, synthetic gauge profiles, laboratory systems that

realize curved-spacetime Hamiltonians—rather than weak-field terrestrial gravimetry.

In developing the architecture, one result organized the design and constitutes the paper's central content. When spacetime curvature is encoded in a dimerized topological chain as position-dependent hopping, the chain's two protected edge modes do not report the curvature equally: one end responds strongly to a warping change while the other is nearly blind—a computed ratio of 151 at the 250-cell reference chain between the two placements of an identical probe on an identical chain under identical curvature, and the law fixes the scaling: the quiet end's response falls inversely with chain length, so the asymmetry grows linearly with the size of the lattice. We derive the law behind this asymmetry: the sensitivity of an edge reading equals the metric gradient integrated over the mode's own probability footprint. Beyond the placement law itself, the derivation yields two exact statements: the zero mode's response to any uniform metric change vanishes identically—not merely to first order—because a uniform rescaling of the bonds leaves the mode's shape unchanged; and the mode's sensitivity admits a closed form as an overlap between its probability footprint and the differential metric tilt, computable in one line from the bond profile. The law is a log-derivative of the standard zero-mode product (Rajbongshi and Modak's exact wavefunctions), and its content is physical rather than deep: the sensitive end of a warped chain is the horizon end, where the local velocity goes to zero, and the quiet end is quiet because the metric is flat there. Placement, not precision, is the first-order design variable.

The instrument that follows from this law is a warped Su-Schrieffer-Heeger chain read by a weakly coupled qubit, whose curvature-dependent decoherence rate is measured not through fragile phase coherence but through the power it dissipates—a calorimetric observable within demonstrated reach of cryogenic bolometry. We present a worked superconducting implementation in which every parameter is derived here, cited to demonstrated hardware, or stated explicitly as a design choice within demonstrated ranges, yielding detectable signals at hundred-second integration for percent-level metric changes.

We believe the placement law is the transferable contribution, the placement asymmetry and its growth with chain length its falsifiable test, and the architecture—particularly in its later array and two-dimensional iterations—a path toward giving the community that studies laboratory spacetime phenomena what it currently lacks: an imager.

2. Background

2.1 The SSH model and its edge states

The Su-Schrieffer-Heeger model describes a one-dimensional chain in which the hopping amplitude alternates between two values, t1 within a unit cell and t2 between cells. Originally introduced to describe the dimerization of polyacetylene, it is the minimal model of a one-dimensional topological insulator. Its defining structural property is chiral (sublattice) symmetry: the Hamiltonian connects sites of sublattice A only to sites of sublattice B. Writing S for the operator that is +1 on A sites and -1 on B sites, the Hamiltonian obeys SHS = -H, which forces the energy spectrum to be symmetric about zero and permits states at exactly zero energy to reside on a single sublattice.

The chain admits two phases distinguished by a winding number: when the inter-cell bond dominates, the winding number is one and each end of a finite chain hosts an exponentially localized zero-energy edge mode; when the intra-cell bond dominates, the winding number is zero and no such mode exists. The transition between the phases requires the bulk gap to close, which is the content of the protection: no deformation of the hopping pattern that preserves chiral symmetry and keeps the gap open can remove the edge mode or move it from zero energy. An intuitive picture is available and exact: in the topological arrangement the chain terminates on a site whose strong bond reaches toward a partner that does not exist. That stranded, unpaired site is the edge mode, and its energetic isolation in the middle of the bulk gap is what makes it so

cleanly addressable.

2.2 The curved-spacetime SSH model

Rajbongshi and Modak [arXiv:2504.16200] introduced spacetime curvature into the SSH model through an equivalence between the Dirac equation in a curved static background and a tight-binding chain with position-dependent hopping, in which the local velocity is set by the local hopping amplitude, v(x) = 2t(x). A Rindler-type metric maps onto a power-law hopping profile, t1(n) = t1 ((2n-1)/(2N-1))^sigma and t2(n) = t2 ((2n)/(2N-1))^sigma, where sigma parameterizes the warping and sigma = 0 recovers the flat chain. Three of their results carry the present work. First, position-dependent hopping of any profile preserves chiral symmetry, so the symmetry class (BDI) and the winding number survive arbitrary warping: curvature deforms the topological phase without destroying it. Second, the zero-energy edge modes of the warped chain become spatially asymmetric between the two ends, in contrast with the mirror-symmetric modes of the flat chain. Third, the zero-mode wavefunctions admit exact expressions as running products of bond ratios, a structure that section 4 converts directly into a sensitivity law. The present work treats the warped chain as a transducer: the metric is imprinted in the bond profile, the edge mode's shape reports on it, and the task is to read that shape out.

2.3 Qubit decoherence as a spectroscopic probe

The readout concept builds on an established line of work in which a qubit weakly coupled to a system serves as a probe of that system's local spectral density. When a two-level system is coupled with strength t_C to a chain at the qubit's transition energy, Fermi's golden rule gives a decoherence rate proportional to t_C^2 multiplied by the local density of states at the coupling site. A qubit probe of SSH edge states on this principle was analyzed in arXiv:1910.09926, which found order-of-five-decade contrast in decoherence rate between chains with and without an edge state at the probed end, together with the ability to distinguish which end hosts the mode. A scanning implementation that maps edge-state profiles by moving the qubit along the chain appears in arXiv:2302.13771. These works establish the probe; they do not consider curved chains, calorimetric readout, or arrays. The contribution claimed here is confined to that combination.

3. Readout Architecture

3.1 The recursive Green's function engine

All spectral quantities in this work are computed from surface Green's functions obtained by decimation. The method deserves a short derivation because it is exact, and its exactness matters to the claims. For a site coupled by hopping t to a semi-infinite remainder whose surface Green's function is g, elimination of the remainder's degrees of freedom replaces it by a self-energy Sigma = t^2 g appended to the site's energy, giving a new surface Green's function g' = 1/(E - eps - t^2 g). No approximation enters: the remainder's equation of motion is solved and substituted, and iterating the substitution site by site is algebraically identical to inverting the full Hamiltonian at a cost linear rather than cubic in chain length. For the semi-infinite uniform lead the recursion closes on itself and yields the standard result g_lead(E) = (E - i sqrt(4t_L^2 - E^2))/(2 t_L^2) within the band; at zero energy the lead self-energy is purely imaginary, so the lead acts as an ideal absorber at its own end. It does not broaden the probed mode: the probed-edge mode's weight at the lead end is of order 10^-151 on the reference chain, so the width of the probed response is set entirely by the broadening parameter of the computation, which stands in for a physical linewidth this draft does not yet design (status note). The recursive and direct-inversion results agree at the level of 10^-15, i.e. to machine precision.

3.2 The calorimetric observable

The probe qubit couples to one end of the warped chain, with the opposite end

terminated by the lead. The qubit decoheres at rate Gamma = 2 pi t_C^2 rho(0), where rho(0) is the local density of states at the coupling site and at the qubit energy, and each decoherence event deposits one quantum, hbar omega, into the dissipative environment. The proposed observable is the resulting power, P = Gamma hbar omega, measured calorimetrically at the cold stage. Two features recommend this choice. It is a classical, integrating measurement that does not require phase-coherent readout of the fragile mode it monitors; and it inherits the full dynamic range of the underlying rate. For a 5 GHz qubit each quantum is 3.3 x 10^-24 J, and $\hbar\omega/kT \approx 16$ at 15 mK, so the thermal occupation of the mode is negligible (e^-16). Section 5 argues the measurement is limited by detector noise rather than by the physics—subject to the status note above, since the absolute rate is not yet fixed.

The coupling strength occupies a genuine tension that should be stated rather than obscured. Back-action of the probe on the chain is of order t_C^2 and must be kept small to avoid deforming the topology being measured; the same t_C^2 sets the signal rate. All numbers quoted in this paper respect the weak-coupling side of that tension, with the probe channel engineered a factor of one hundred above the qubit's intrinsic decoherence background so that the chain-induced rate dominates the power budget while remaining far below any scale that would perturb the mode structure.

3.3 Contrast with the physical Hamiltonian

Using the power-law profile of section 2.2 rather than any linearized surrogate, the computed zero-energy surface density of states at the probed end differs between the topological and trivial arrangements by 3 x 10^11 at N = 250 cells and broadening 10^-7 (the ratio scales as the inverse square of the broadening and is therefore set in a physical device by the physical linewidths rather than by this numerical figure). The edge state is therefore not a subtle feature to be extracted from a background; it is a binary beacon, and the measurement problem is entirely about resolving changes in its shape, which is the subject of the next section.

4. Sensitivity Theory: the Placement Law

4.1 The zero mode as a running product

Chiral symmetry permits a zero-energy solution confined to one sublattice. Imposing H psi = 0 on such a state yields one constraint per B site, t1(n) a_n + t2(n) a_{n+1} = 0, whose solution is a running product: a_n = a_1 prod_k (-t1(k)/t2(k)). When the bond ratios are less than one in magnitude the amplitude decays geometrically from the edge, and the squared amplitudes w(n) = |a_n/a_1|^2 constitute the mode's probability footprint. This product structure is exact and matches the closed-form wavefunctions of the curved-chain reference.

4.2 Derivation of the sensitivity law

Let curvature perturb every bond multiplicatively, t(k) -> t(k)(1 + eps f(k)), with eps small and f the metric profile. Because the mode is a product of bond ratios, its logarithm is a sum, and the perturbation enters as ln w(n) = [flat part] + 2 eps Phi(n), where Phi(n) is the accumulated differential tilt, the sum of f1(k) - f2(k) from the edge out to cell n. The probed quantity is the normalized edge weight, rho_edge proportional to 1/sum_n w(n), whose logarithmic derivative follows in two lines:

d ln rho_edge / d eps = -2 ,

the average of the accumulated tilt weighted by the mode's own probability footprint. This is the placement law: the sensitivity of the edge reading to curvature equals the metric's differential variation integrated over the region the mode actually occupies.

4.3 Two corollaries

The first corollary is exact invisibility of uniform metric changes. If both bond species are stretched identically at every position, the differential tilt vanishes term by term, Phi(n) = 0, and the zero-mode reading is unchanged not merely to first order but

identically. This is scale invariance, not more: H → λH leaves every eigenvector unchanged, and E = 0 is distinguished only as the fixed point of E → λE, so the zero mode's shape — and the edge weight read by a probe at zero energy — cannot see λ, while finite-energy observables are strongly scale-sensitive. The statement also depends on the configuration of the reference: a uniform rescaling that includes the probe coupling is invisible for the independent reason that the probe is then comoving with the system, whereas a rescaling that excludes a fixed probe coupling is observable, which is the lattice analogue of a redshift measured against an external standard.

The second corollary is that placement decides the magnitude of the response. The weighted average is large only where the mode's footprint overlaps the region in which the metric profile varies steeply. The power-law warping varies steeply near the coordinate origin and plateaus far from it; consequently a mode localized at the near end reports strongly while a mode localized at the far end reports almost nothing. Computed at sigma = 0.5 on the 250-cell reference chain, the logarithmic responsivities of the two ends are +0.203 and -0.00134: a factor of 151 between correct and incorrect placement of the same probe on the same chain under the same curvature, with each edge's value verified against the per-edge form of the law to four digits at broadening 10^-5 (at 10^-3 the horizon-end value drifts by 0.1 percent, the near-horizon gap approaching the broadening). The physical content is that the sensitive end is the horizon end: in Rajbongshi and Modak's metric the local velocity vanishes there, the bonds go soft, and the mode's footprint sits where the profile is steepest; the far end sits on the flat plateau. The law makes the quiet end quantitative: on the plateau the accumulated tilt after j cells is ≈ j/(2N), the footprint average of j is r^2/(1 − r^2) = 1/3 at r = 0.5, so the quiet-edge responsivity is ≈ −1/(3N) — −0.00133 at N = 250 against the computed −0.00134 — and the asymmetry grows linearly with chain length: computed 60-fold, 151-fold, and 304-fold at N = 100, 250, and 500. This scaling follows from the parametrization x ∈ (0, 1] stretched over N cells, and the prediction should be read as such: a test of the law's footprint structure, not a surprise about the chain. This asymmetry and its scaling law are the paper's principal falsifiable prediction, testable in any platform that can realize warped dimerized chains of two or more lengths and a movable weakly coupled probe, including room-temperature photonic arrays.

4.4 Design rule

The corollaries compress into a single instruction: match the probe location and the mode's footprint to the steepest region of the metric being imaged. Localization depth is not by itself desirable or undesirable; overlap with the gradient is the entire figure of merit. Softening the dimerization toward the topological transition is one lever for that matching: delocalizing the mode extends its footprint into regions where the power-law profile still varies. The computed fractional responsivity rises from +0.20 at r = 0.50 to +2.26 at r = 0.97 under the same warping (broadening-converged; the value is stable across eta = 10^-5 to 10^-7 and chain lengths 250 to 500, and an earlier lead-terminated computation gave +2.35). The order-of-magnitude gain is purchased at a 3.6-fold reduction in absolute signal, with a null contrast that remains above 10^10. We note for the record that an earlier exploration under a linearized warping profile had indicated no such gain; the finding did not survive the transfer to the physical power-law profile, and only the power-law result is claimed here. The operating point is therefore a genuine optimization between responsivity and absolute signal, set by detector considerations; the conservative deep-phase point r = 0.50 is used for the worked example of section 5, with near-critical operation available as computed upside.

5. Worked Example: Superconducting Implementation

The superconducting circuit platform is selected because it is, to our knowledge, the only ecosystem in which the three necessary ingredients coexist natively: decoherence rates in the megahertz range, an engineered dissipative environment with a literal heatsink at the mixing-chamber stage, and demonstrated calorimetry at the required noise floor within the same cryostat and frequency band. SSH physics has been

realized in superconducting arrays, with cell couplings implemented capacitively or inductively and demonstrated coupling strengths spanning 10 to 250 MHz; position-dependent hopping is in this platform a lithographic instruction rather than a dynamical engineering problem, and tunable couplers permit in-situ trimming.

The reference design takes a transmon-class probe at 5 GHz coupled at t_C in the low-megahertz range to the near end of a warped chain, the far end terminated in an engineered cold load. With intrinsic probe coherence of 100 microseconds, mid-range among reported values, the chain-induced channel at Gamma = 10^6 per second dominates the intrinsic background by two orders of magnitude, so that 99 percent of the dissipated power carries signal. The dissipated power is P = Gamma hbar omega = 3.3 attowatts. Against this, the demonstrated bolometer noise floor of 30 zeptowatts per root hertz [Nature 586, 47 (2020) class] defines the measurement resolution; the photon shot noise of the dissipation itself lies an order of magnitude below the detector floor, confirming detector-limited operation.

A change δσ = 0.01 in the warping exponent (a position-dependent bond change of up to ~6 percent near the horizon end) at the correctly placed end produces, through the responsivity of section 4, a fractional power change of 2 x 10^-3, i.e. 6.6 zeptowatts of the 3.3 attowatts assumed — figures whose absolute scale is under revision per the status note. The resulting signal-to-noise ratio is approximately 2 at 100 seconds of integration and approximately 22 at three hours. These figures are quoted at the conservative deep-phase operating point. The near-critical operation computed in section 4.4 offers up to an order of magnitude in additional responsivity at modest signal cost; it is left as characterized upside rather than folded into the quoted figures. These figures are slow by the standards of transport spectroscopy and entirely honest: every number in the chain is derived above, cited to demonstrated hardware, or flagged as an explicit design choice, no differential rejection is assumed, and the detector-limited character of the result means the integration times shorten quadratically with future improvements in calorimetry. Thermal occupation imposes no near-term ceiling, with $\hbar\omega/kT \approx 16$ at the operating temperature.

Parameter	Value	Basis
Probe frequency f	5 GHz	standard transmon-class
Energy per event $\hbar\omega$	3.3×10^{-24} J	derived
Demonstrated couplings	10–250 MHz	cited (superconducting SSH realizations)
Intrinsic coherence T_1	100 µs	cited mid-range (10 µs–1 ms reported)
Engineered probe rate Γ	10^6 s^{-1}	design choice within cited ranges
Signal power P	3.3 aW	derived
Intrinsic background	1% of P	derived from T_1
Operating temperature	15 mK	standard dilution stage
$\hbar\omega/kT$	16	derived
Detector NEP	30 zW/√Hz	demonstrated (Nature 586, 47 (2020) class)
Shot-noise floor	~3 zW at 1 s	derived; 10× below detector
Responsivity (correct placement)	d ln ρ/dσ ≈ +0.20	computed, σ = 0.5
Responsivity (wrong placement)	≈ −0.001	computed
Signal, δσ = 1%	6.6 zW	derived
SNR at 100 s / 3 h	≈ 2 / ≈ 22	derived

6. Toward Differential Array Imaging

The single element of section 5 is the pixel; the instrument promised in the introduction is the array. This section states what the array adds and what it awaits.

What it adds is the frame-independent object. A single element reads one projection of

the local metric gradient. An array, differenced between neighbors, reads the gradient of the gradient—the tidal quantity that no choice of frame can remove. In this sense the array is a lattice of test masses performing a geodesic-deviation measurement, and differencing additionally rejects perturbations common to neighboring elements.

What it awaits is two determinations. The first is the noise correlation structure across a shared substrate: common-mode noise favors differencing, independent noise favors averaging, and which regime the hardware occupies is an empirical question not settled here. No rejection factor is assumed anywhere in this paper's quoted sensitivities, which are single-ended throughout. The second is calibration stability. Fabrication scatter at the percent level exceeds the neighbor-to-neighbor tidal signal by roughly two orders of magnitude; the scatter is static, and therefore removable against a flat-metric reference, but the residual specification then falls on drift between calibrations, which remains an open engineering requirement.

7. Validation Path

The claims of this paper admit a staged validation, ordered by cost, and the stages are separable because they test different layers of the architecture. The first stage is photonic and requires neither cryogenics nor quantum hardware. A warped dimerized waveguide array—coupling set by waveguide spacing, the power-law profile written directly into the geometry—realizes the curved-SSH Hamiltonian at room temperature, and a weakly coupled probe waveguide translated along the array samples the local mode weight at either end. The placement law's central prediction—the placement asymmetry, a factor of 151 at the 250-cell reference chain, and its linear growth with chain length—is testable in this platform for the cost of a fabrication run. A series of such arrays spanning dimerization ratios would additionally trace the responsivity-versus-r curve of section 4.4, converting the falsification target from a single ratio into an entire predicted functional form. A null result here falsifies the paper's core claim before any cryogenic investment; a positive result validates the placement law and the exact uniform-cancellation corollary, though not the calorimetric chain, since classical light dissipating in a lossy array pays no coherence-maintenance cost.

The second stage is atomic. Cold atoms in an optical lattice with modulated hopping realize the warped chain with dynamical control of the profile—the route the model's own authors identify—and give access to Hamiltonian-level physics the photonic stage cannot reach: the asymmetric zero-mode structure under genuine quantum statistics and the adiabatic response of the mode to a time-varying warping. This stage validates the transducer. It does not validate the readout, because the atomic platform's dissipation channels (spontaneous emission, atom loss) do not constitute a measurable coherence-maintenance power in the sense this paper's observable requires.

The third stage is the instrument itself: a superconducting chain of modest length, its bond profile set lithographically, one end terminated in an engineered cold load, a transmon-class probe coupled to the other, and a cryogenic bolometer integrating the dissipated power. The minimal experiment is not an image but a single differential measurement: the dissipated power at the two probe placements, under the same fabricated warping, compared against the placement law's quantitative prediction. That one number exercises every link in the chain this paper computes—the warped Hamiltonian, the mode asymmetry, the golden-rule rate, the calorimetric observable, and the detector budget—and its agreement or disagreement with section 5 is the direct experimental verdict on the parts of this work that currently rest on our numerics alone.

8. Limitations and Honest Scope

This architecture began as an ambitious push toward a general spacetime sensor, and the mathematics disciplined that ambition in three ways, which we report as findings rather than footnotes. First, this instrument cannot compete with optical atomic clocks on absolute time dilation, and the margin is not close: matching demonstrated clock stability of 10^-18 would require of order 10^27 array elements at our correction rates

—a number that ends the discussion. Absolute dilation metrology belongs to clocks, and we concede it entirely. Second, the project's original framing targeted femtosecond time resolution, and that framing was simply wrong: it misidentified the figure of merit. What governs a dilation measurement is fractional frequency stability, where clocks are supreme; what this architecture offers instead is spatial resolution—many matched elements imaging how a metric varies across a region—which is a different quantity serving a different question. Third, the design carries a hard limit of its own making: the topological protection that stabilizes the probe mode and gives the readout its enormous contrast is the same property that suppresses the mode's response to perturbation. Protection and sensitivity draw from the same account. The placement law of section 4 is how the design pays that bill—by spending the mode's limited responsiveness exactly where the metric gradient is steepest—but the tension is structural, not an engineering artifact to be optimized away.

Within that disciplined scope, the claims divide cleanly into what stands on published ground and what stands, for now, on our numerics alone, and we draw that line explicitly. The foundation is validated: the curved-spacetime SSH Hamiltonian, the survival of the topological phase under warping, and the asymmetric structure of the zero modes are results of Rajbongshi and Modak, which our independent calculations reproduce, and the qubit-decoherence probe of SSH edge states is established prior art. What is ours—and, as of the cold review of October 2026, examined by one outside reader whose findings are recorded in the repository—is the sensing chain built on that foundation: the calorimetric observable, the power-dissipation readout, and the placement law's quantitative predictions are computed results awaiting experimental validation. That chain is exercised, moreover, under a single metric family, the power-law Rindler-type profile; its behavior under other curvature profiles is unestablished. The chain is also tested only against hopping disorder, which the protected mode is immune to by construction; on-site (frequency) disorder — the kind superconducting elements actually have — moves the mode's energy by roughly 0.6 of the scatter while leaving its shape and the laws intact, and implies an element-frequency trimming specification that this draft does not yet state. And the 10^6 per second probe rate that anchors the worked example is a plausible engineered value chosen within demonstrated coupling ranges, not a rate that has been realized and measured in a warped-chain device. We flag these not as hedges but as the specific experiments this paper is asking for. Finally, the contribution should be fenced accurately: a single qubit probing a topological chain is not new. What is proposed here is a pathway—curvature encoded in the bond profile, read calorimetrically, optimized by the placement law, and extending through array differencing toward imaging—of which this paper establishes the first stage.

Where the physics is ours, the performance costs are stated with equal candor. The first cost is time. At the demonstrated bolometer floor, a one-percent metric change yields a signal-to-noise ratio of approximately 2 over one hundred seconds of integration and approximately 22 over three hours: glacial by the standards of transport measurement, with quality purchased only by patience. The measurement is detector-limited—the photon shot noise of the dissipation itself sits an order of magnitude below the bolometer floor—which is the favorable failure mode, since improvements in cryogenic calorimetry convert directly into integration time. But we state the boundary of that claim precisely: our noise budget is computed up to the detector, and it identified the detector as the limit; the amplifier chain and readout-line contributions beyond the bolometer are not yet modeled and remain further work. The second cost is delicacy. Fabrication scatter at the percent level stands roughly two orders of magnitude above the neighbor-to-neighbor tidal signal, so the instrument is calibration-dependent by construction: the scatter is static and removable against a flat-metric reference, whereupon the operative specification migrates to drift stability between calibrations, which we have not quantified.

These constraints do not scatter in independent directions; they converge on a single program of further work. The single element characterized here becomes genuinely useful as an array, and the array's viability turns on questions this paper deliberately leaves open: the correlation structure of noise across a shared substrate, which decides whether averaging or common-mode rejection is the winning strategy; the element

footprint, which sets the pixel size and dominates reconstruction error well before element mismatch does; and the physical mechanism by which the warping profile is written onto hardware, for which tunable couplers are the identified candidate but no design is offered here. Beyond the array lie the two-dimensional extension, whose higher-order corner modes promise sharper probes and access to the full tidal tensor, and a companion line of work in which the same chains, operated in a different parameter regime, serve as protected cryogenic memory elements co-located with the imager. Those developments belong to subsequent papers. This one claims the one-dimensional placement physics: the law, its two exact corollaries, and the placement asymmetry — 151-fold at the reference chain, linear in chain length — that any laboratory with a warped dimerized chain and a movable probe can test.

9. Conclusion

The transferable result of this work is a law, not a device. The sensitivity of a topological edge mode to spacetime curvature encoded in its lattice equals the differential metric tilt integrated over the mode's own probability footprint: uniform changes cancel exactly by scale invariance, only gradients register, and placement of the probe against the gradient—not the precision of the probe—is the first-order design variable. The law converts a question that would otherwise be settled by trial fabrication into a one-line calculation from the bond profile, and it carries a falsifiable signature: a placement asymmetry exceeding two orders of magnitude and growing linearly with chain length, testable at room temperature in a photonic array. Around that law we have specified an instrument—a calorimetric readout of the coherence-maintenance power of a warped superconducting chain—whose responsivities are derived here and whose absolute power awaits the design of its dissipative channel (status note), and whose honest performance, slow and calibration-dependent, defines a niche no existing instrument occupies: not the measurement of the metric at a point, at which optical clocks are beyond reach, but the imaging of how an engineered metric varies across a region. The single element presented here is the first stage of that program; the array, the two-dimensional extension, the co-located memory, and the stationary-metric channel — the shift-sensing extension developed in a companion paper, where the placement law acquires an exact quadratic analog — are its continuation.

Appendix A: Numerical Methods

Surface Green's functions were computed by the decimation recursion of section 3.1 with a positive infinitesimal broadening eta between 10^-3 and 10^-7 in units of t2 (stated per result), verified against direct matrix inversion to machine precision at all parameters reported. Zero-energy local densities of states follow as -(1/pi) Im G^S(0). Responsivities were obtained by symmetric finite differences in the warping parameter with step 10^-4 to 10^-5, checked for step-size independence. The curvature profile is the power-law form of section 2.2 throughout; earlier exploratory results obtained with a linearized profile were retained only where reconfirmed under the power-law form. Chain lengths of 100 to 500 cells were used as noted; the qualitative structure (edge contrast, placement asymmetry, uniform invisibility of the zero mode) is insensitive to length above roughly 40 cells, while the asymmetry's magnitude grows linearly with chain length as quantified in section 4.3. The placement law was verified directly against finite-difference numerics: on the power-law family, on generic smooth random metric perturbations, and at each edge of the even chain against the per-edge form of the law, with agreement of 0.1 percent or better in typical regimes and a few percent where soft local gaps approach the broadening scale, consistent with background contamination; the quiet edge's responsivity is broadening-independent, confirming mode physics rather than background. Near-critical responsivities are stable to a few percent across broadening 10^-3 to 10^-7 and chain lengths 250 to 500, converging to +2.26 at r = 0.97; the accompanying 3.6-fold reduction in absolute edge density of states is printed by the same script. Scripts: the persisted verification set comprises curvature_law_check.py and asym_check.py (direct law-vs-numerics verification, both

edges, generic profiles, N-scaling), hole_check.py (near-critical stability in broadening and length), and recursion_check.py (decimation vs direct inversion, agreement 4 x 10^-16). Earlier exploratory scripts (ssh_gf.py, honest_rerun.py, leftedge.py, nearcrit_powerlaw.py, worked_example.py, imager.py, ssh_memory_unified.py) were session-local; every headline figure they produced has been independently re-verified by the persisted set or is reproducible from the formulas of this appendix and the companion derivations document.

References

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